Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel
Level 1/Level 2 GCSE (9–1)

Hannah Kettle MATHS TUTORIALS

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Half Past Paper – Week 21

Higher Tier

	Total Marks

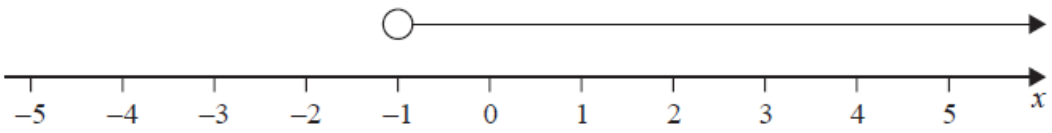
Q	Topic	Marks	Sparx Code
1a	Reading and drawing inequalities on number lines	1	U509
1b	Reading and drawing inequalities on number lines	2	U509
3a	Calculating speed from distance-time graphs	2	U462
3b	Plotting distance-time graphs using speeds	3	U966
5	Using Pythagoras' theorem in 2D	4	U385
7	Calculating with density	3	U910
9a	Interpreting box plots	2	U837
9b	Interpreting box plots	1	U837
9c	Comparing populations using box plots and cumulative frequency graphs	1	U507
11a	Solving shape problems involving coordinates	2	U889
11b	Equations of parallel and perpendicular lines	3	U898
13	Using the product rule for counting	2	U369
15a	The cosine rule	3	U591
15b	The area rule	2	U592
17	Interpreting histograms	4	U983
19	Finding the surface area and volume of similar shapes	3	U110
21	Changing the subjects of formulae, Finding error intervals for calculations	4	U556,U587
	Total	42	

Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

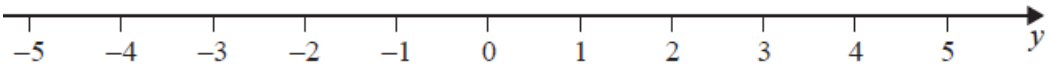
1 (*a*) Write down the inequality shown on this number line.



.....

(1)

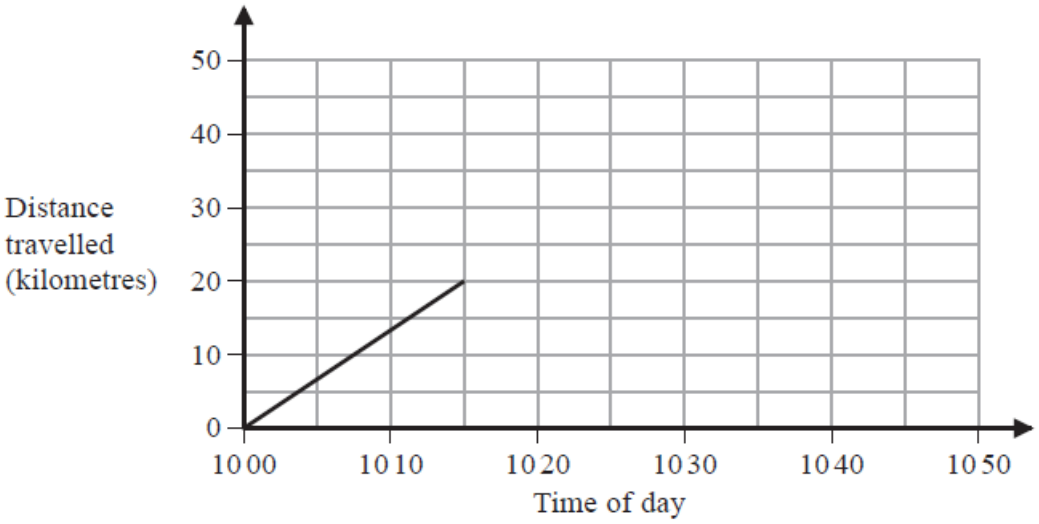
(*b*) On the number line below, show the inequality $-3 \leq y < 4$



(2)

(Total for Question 1 is 3 marks)

3 Sam drives his car on a journey.
Here is the travel graph for the first 15 minutes of his journey.



(*a*) Work out Sam's speed, in km/h, for the first 15 minutes of his journey.

…..... km/h
(2)

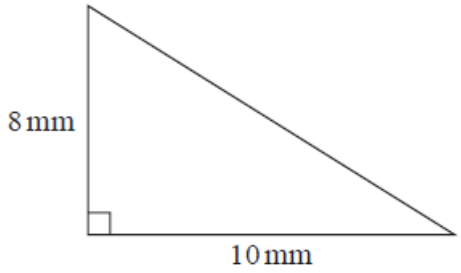
At 10 15 Sam stops for 10 minutes and then drives for 20 minutes at a speed of 75 km/h.

(*b*) On the grid, complete the travel graph for Sam's journey.

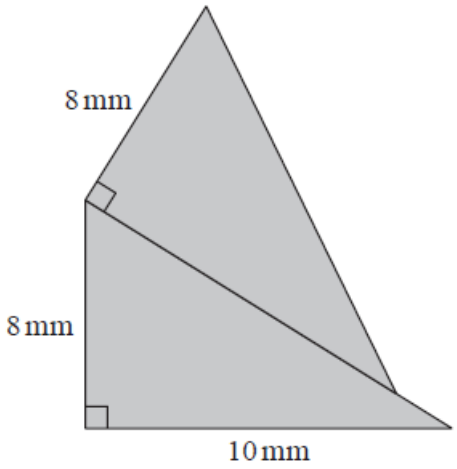
(3)

(Total for Question 3 is 5 marks)

5 Here is a right-angled triangle.



The shaded shape below is made from two of these triangles.



Work out the perimeter of the shaded shape.
Give your answer correct to 3 significant figures.

..... mm

(Total for Question 5 is 4 marks)

7 Liquid **A** has a density of 1.8 g/cm^3
Liquid **B** has a density of 1.2 g/cm^3

80 cm^3 of liquid **A** is mixed with 40 cm^3 of liquid **B** to make 120 cm^3 of liquid **C**.

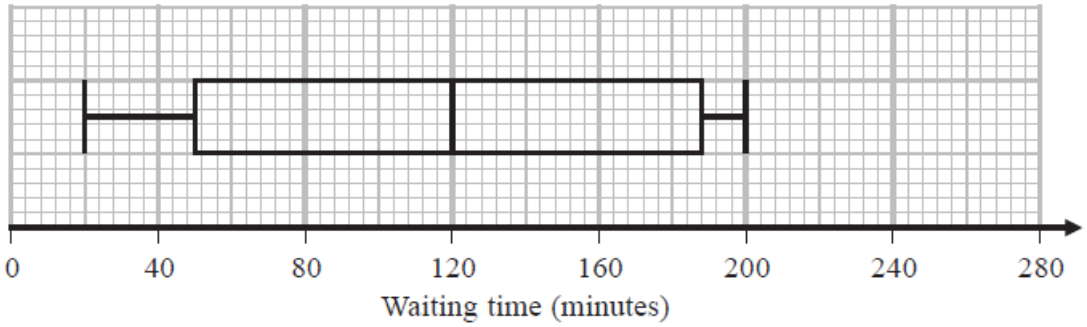
Work out the density of liquid **C**.

..... g/cm^3

(Total for Question 7 is 3 marks)

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9 The box plot shows information about the length of time, in minutes, some people waited to see a doctor at a hospital on Monday.



(*a*) Work out the interquartile range of the information in the box plot.

..... minutes

(2)

Becky says,

"50% of the people waited for at least 2 hours."

(*b*) Is Becky correct?
Explain why.

.....

.....

.....

(1)

The table gives information about the length of time, in minutes, some people waited to see a doctor at the same hospital on Tuesday.

	Waiting time (minutes)
Shortest time	20
Lower quartile	50
Median	100
Upper quartile	140
Longest time	210

Becky was asked to compare the distribution of the lengths of times people waited on Monday with the distribution of the lengths of times people waited on Tuesday.
She wrote,

"People had to wait longer on Tuesday than on Monday."

(*c*) Give **one** reason why Becky may be wrong.

.....

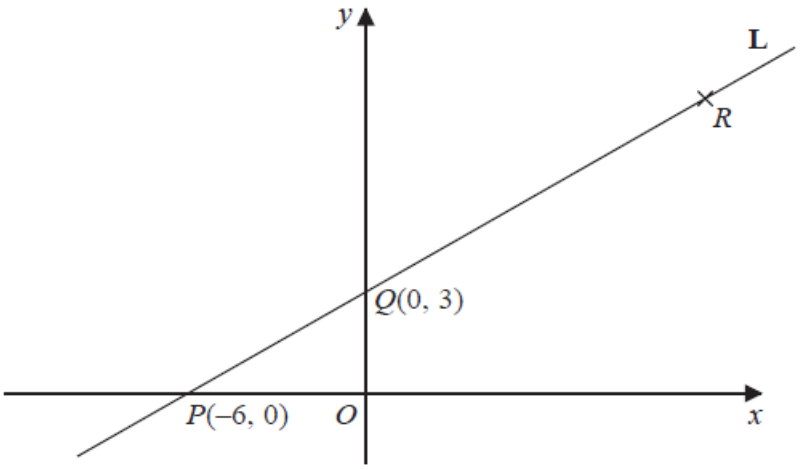
.....

.....

(1)

(Total for Question 9 is 4 marks)

11 Here is a sketch of the line **L**.



The points $P(-6, 0)$ and $Q(0, 3)$ are points on the line **L**.

The point R is such that PQR is a straight line and $PQ : QR = 2 : 3$

(*a*) Find the coordinates of R.

(..... , ...)

(2)

(*b*) Find an equation of the line that is perpendicular to **L** and passes through Q.

.....

(3)

(Total for Question 11 is 5 marks)

13 In a school there are 16 teachers and 220 students.
Of these students 120 are girls and 100 are boys.

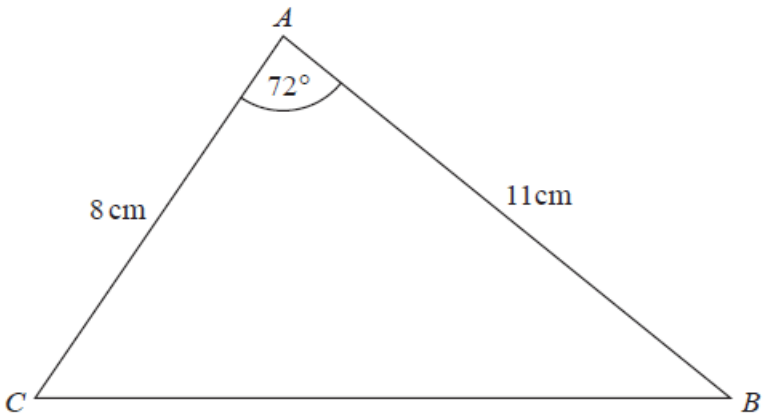
One teacher, one girl and one boy are going to be chosen to represent the school.

Work out the number of different ways there are to choose one teacher, one girl and one boy.

.....

(Total for Question 13 is 2 marks)

15 Here is triangle ABC.



(a) Find the length of BC.
Give your answer correct to 3 significant figures.

..... cm
(3)

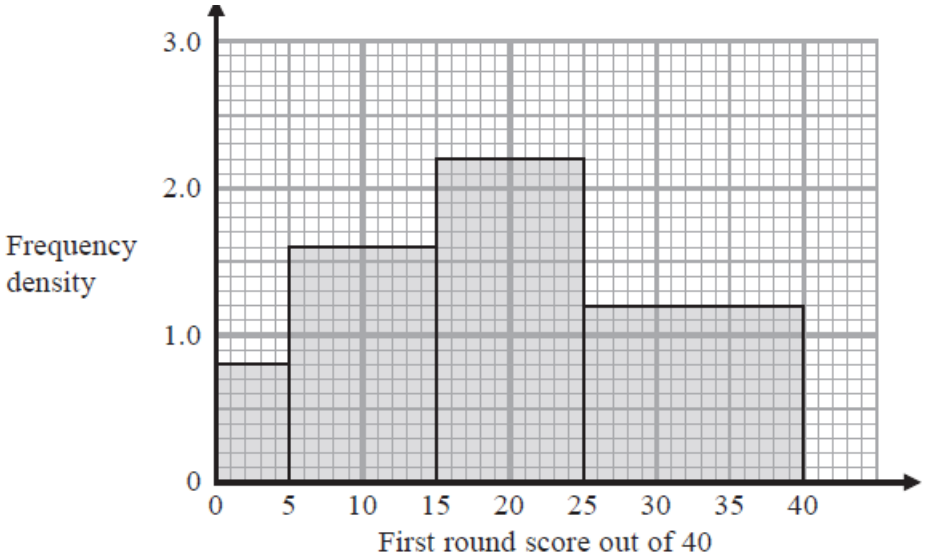
(b) Find the area of triangle ABC.
Give your answer correct to 3 significant figures.

..... cm^2
(2)

(Total for Question 15 is 5 marks)

17 Some people took part in the first round of a competition.

The histogram gives information about the scores of these people in the first round.



20% of the people got a score high enough for them to qualify for the second round.

Work out an estimate for the score needed to qualify for the second round.
You must show all your working.

.....

(Total for Question 17 is 4 marks)

19 **A**, **B** and **C** are three spheres.

The volume of sphere **A** is 125 cm^3
The volume of sphere **B** is 27 cm^3

The ratio of the radius of sphere **B** to the radius of sphere **C** is 1 : 2

Work out the ratio of the surface area of sphere **A** to the surface area of sphere **C**.

.....

(Total for Question 19 is 3 marks)

21 The time period, T seconds, of a simple pendulum of length l cm is given by the formula

$$T = 2\pi\sqrt{\frac{l}{g}}$$

Katie uses a simple pendulum in an experiment to find an estimate for the value of g.

Here are her results.

$l = 52.0$ correct to 3 significant figures.
$T = 1.45$ correct to 3 significant figures.

Work out the upper bound and the lower bound for the value of g.
Use $\pi = 3.142$
You must show all your working.

upper bound = ...

lower bound = ...

(Total for Question 21 is 4 marks)

TOTAL FOR PAPER IS 42 MARKS